

Automation, AI, And Labor Demand

Displacement, Augmentation, Scale, And New Tasks

Special Topic 5: Technology, Innovation, and Labor – Week 2

Central question

- Which tasks and workers are displaced when automation or AI changes production?
- Which workers are augmented because technology raises productivity in their remaining tasks?
- When do productivity and output-demand effects offset displacement?
- When does new-task creation move labor demand into different work?

Labor-demand object

- The object is not technology in the abstract.
- The object is changing demand for tasks, skills, occupations, establishments, firms, industries, and local labor markets.
- A good estimate names the technology margin, the exposure mapping, the labor outcome, and the adjustment horizon.
- Week 2 discipline: treat measurement as part of the theory.

Four demand channels

Channel	Task effect	Labor-demand question
Displacement	labor tasks move to capital/software	whose tasks shrink?
Augmentation	workers become more productive	whose marginal product rises?
Output/scale	productivity expands sales/output	does hiring recover elsewhere?
New tasks	new activities or occupations appear	where is labor reinstated?

$$\Delta L = \Delta L^{disp} + \Delta L^{aug} + \Delta L^{scale} + \Delta L^{new}$$

Exposure is a mapping

$$Exposure_o = \sum_{k \in K} s_{ok} \theta_k$$

- s_{ok} is the task share of task k in occupation o .
- θ_k is the technology's effect on the cost or productivity of task k .
- Exposure predicts susceptibility; it is not adoption and not an effect.
- The outcome and counterfactual determine the labor-demand interpretation.

Unit of incidence

Unit	What the design sees	Main caution
Occupation	task bundles and exposure	not adoption
Firm/establishment	adoption, reorganization, hiring	adopter selection
Industry	sector productivity and input mix	masks workers
Local market	equilibrium incidence across places	bundled shocks

- A technology can displace a task, raise employment at adopters, and lower wages in exposed places.

Why wages and employment diverge

- Direct task displacement can lower employment in exposed work.
- Surviving jobs can become more skill-intensive, raising average wages.
- Productivity can lower prices and expand output demand.
- Adoption can shift rents between workers, firms, and consumers.
- Local labor markets adjust through mobility, entry, exit, vacancies, and occupational change.

Measurement menu

Measure	Closest object	Main warning
Robot data	embodied automation adoption	narrow technology class
Task exposure	susceptibility of job content	not realized use
AI exposure	capability/task match	adoption frictions matter
Patents/text	invention direction	far from implementation
Firm adoption	realized organizational use	strong selection
Vacancy text	desired skill demand	not employment
Worker-use data	actual tool use/productivity	narrow horizon

Historical comparison

Episode	Labor-demand lesson	AI comparison
Mechanization/steam	displacement and productivity co-exist	direct substitution still needs scale accounting
Electrification	gains require organizational re-design	AI adoption is workflow redesign, not just tool access
Computerization/ICT	routine tasks fall; abstract tasks rise	AI shifts the cognitive boundary of automation
Robotics	embodied automation gives local incidence estimates	AI data are faster but often less concrete
Occupational churn	new work appears slowly and unevenly	short-run AI effects miss new-task creation

What may be distinctive about AI

- It reaches some cognitive, language, coding, prediction, and coordination tasks.
- Diffusion can be fast where deployment costs and governance barriers are low.
- The treatment often works through reorganization rather than one-for-one replacement.
- Measurement can use text, vacancies, prompts, usage logs, and worker productivity data.
- General-equilibrium effects are uncertain because many sectors may adjust at once.

What is not new

- Displacement anxiety has appeared in earlier technology waves.
- Complementary investments and adoption lags are recurring features.
- Skill demand changes through tasks and organization.
- New tasks and occupations can offset displacement, but not instantly or evenly.
- Incidence depends on workers, firms, places, institutions, and adjustment frictions.

Research design checklist

- ① Name the technology object: robot, software, AI capability, AI use, vacancy demand, or patent text.
- ② State the exposure mapping: task, occupation, firm, industry, or place.
- ③ Choose the outcome: employment, wages, vacancies, sales, mobility, task mix, or job quality.
- ④ State the horizon: immediate use, adoption, local adjustment, or long-run churn.
- ⑤ Identify the equilibrium channel left outside the estimate.

- Reproduce: construct a synthetic commuting-zone robot exposure measure.
- Diagnose: compare employment and wage incidence across exposed local markets.
- Transfer: compare local robot exposure with firm AI adoption and vacancy AI demand.
- Goal: learn why local exposure, firm adoption, and worker use are different empirical objects.

- Week 2 asks where labor demand moves when technology changes tasks.
- Week 3 asks how workers adjust once demand has moved.
- The next objects are skills, training, mobility, career transitions, bargaining, and welfare costs.

Takeaways

- Automation and AI operate through displacement, augmentation, scale, and new tasks.
- Employment and wages diverge because incidence is multi-channel and multi-level.
- Historical comparison places AI inside a longer labor-demand literature.
- Measurement choices identify different margins; they are not technical afterthoughts.